SECTION A – Multiple Choice Questions (MCQ)

Instruction:
Each question describes an extension to an existing software project. Choose the **MOST appropriate Design Pattern** to address the requirement.

Q1

A university timetable system needs to ensure that only one instance of the system configuration exists throughout the application runtime to avoid inconsistent scheduling data.

A. Factory
B. Builder
C. Singleton
D. Prototype

Q2

A payment system is extended to support multiple payment methods (Credit Card, E-Wallet, Bank Transfer). The system must decide at runtime which payment object to create based on user selection.

A. Abstract Factory
B. Factory
C. Prototype
D. Builder

Q3

An e-commerce platform is expanded to support different families of UI components (Web UI, Mobile UI). Each family must produce related components such as Button, Form, and Menu that are compatible with each other.

A. Factory
B. Builder
C. Prototype
D. Abstract Factory

Q4

A report generation module now allows reports to be constructed step-by-step, where the same construction process can generate either a PDF or HTML report.

A. Factory
B. Builder
C. Singleton
D. Adapter

Q5

A game development system needs to create multiple enemy objects by cloning existing enemy templates instead of creating them from scratch to improve performance.

A. Factory
B. Builder
C. Prototype
D. Singleton

Q6

A file system application represents folders and files in a hierarchical structure. The extension requires treating individual files and folders uniformly when calculating total storage size.

A. Adapter
B. Facade
C. Composite
D. Iterator

Q7

A legacy weather service provides data in XML format. A new analytics module expects weather data in JSON. The system must integrate the legacy service without modifying its code.

A. Facade
B. Composite
C. Adapter
D. Observer

Q8

A complex subsystem consists of authentication, logging, and authorization modules. A simplified interface is required so that client applications can access these services through a single entry point.

A. Adapter
B. Facade
C. Composite
D. Builder

Q9

A digital library system allows users to traverse a collection of books in different ways (by title, author, or publication year) without exposing the internal structure of the collection.

A. Observer
B. Composite
C. Iterator
D. Prototype

Q10

A smart home system is extended so that multiple mobile apps receive automatic notifications whenever a device’s state changes (e.g., door opened, light turned off).

A. Iterator
B. Composite
C. Facade
D. Observer

SECTION B – Structured / Design Question

Question

A **Smart Campus Management System** is being extended with the following new requirements:

1. The system must support **different families of devices** (Classroom Devices, Laboratory Devices) where each family produces related components.
2. Buildings and rooms must be represented in a **hierarchical structure**, allowing operations to be performed on individual rooms or entire buildings uniformly.
3. When energy consumption changes in any room, **multiple monitoring dashboards** must be automatically notified.

Task

Using **ONE design pattern from EACH category**, answer the following:

a) Identify **ONE Creational Pattern**, **ONE Structural Pattern**, and **ONE Behavioral Pattern** suitable for the system extension.
b) Draw the **UML class diagram** for each selected pattern.
c) Briefly explain how each pattern solves the stated requirement.